

# **CSE3443 SOFTWARE MAINTENANCE AND EVOLUTION**

## **Semester II 2025/2026**

### **Group Project Guidelines**

#### **1. INTRODUCTION**

Software maintenance is a critical aspect of software engineering that involves modifying and updating a software system after it has been delivered to the customer. In this assignment, students will explore the key principles and concepts related to software maintenance, including but not limited to bug fixing, code refactoring, version control, and software documentation. Students will analyze how these principles apply to real-world scenarios encountered in their final year projects, identifying challenges and proposing feasible solutions ensure the long-term sustainability and usability of the software product.

#### **2. INSTRUCTIONS**

##### **Group Member Responsibilities**

- Students must form a group consisting of four (4) members. Each group must select one (1) topic from the list of topics provided by the lecturer. This topic will serve as the main theme of the group report.
- Although the assignment will be submitted as a single group report, the analysis and discussion are conducted individually.
- After selecting the main topic, each group member must derive one specific subtopic related to the chosen topic. The subtopic should focus on a particular aspect of software maintenance and evolution.

##### **Selection of Technical Papers**

Each student must select two (2) technical papers related to their assigned subtopic. The selected papers must meet the following criteria:

- Published within the last five (5) years
- Obtained from credible academic sources, such as: ACM Digital Library, IEEE Xplore, ScienceDirect, Google Scholar
- The selected papers must be unique between groups.

## **Selection of PITA Project**

- Each group must identify one (1) Projek Ilmiah Tahun Akhir (PITA) that aligns with the selected topic.
- The chosen PITA project will serve as the common case study for the entire report.
- All group members must relate their technical paper discussions and analysis to the development practices of the same PITA project.

## **Report Preparation**

Students are required to prepare a detailed report that integrates the individual analyses into one coherent group report.

The report should cover the following components.

### **1. Introduction (Group Section)**

This section should include:

- Background of software maintenance and evolution
- Explanation of the selected main topic
- Introduction to the selected PITA project
- Brief explanation of the subtopics assigned to each group member

### **2. Individual Technical Paper Reviews**

Each student must prepare an individual subsection based on their assigned subtopic.

For each paper, the review should include:

- Research problem
- Methodology
- Key findings
- Strengths and limitations

### **3. Comparison with PITA Development Practices**

Each student must relate the ideas and approaches from the technical papers to the selected PITA project.

The analysis should include:

- Comparison between the methods proposed in the papers and the development practices used in the PITA project

- Discussion on how the techniques could improve the maintainability and evolution of the system
- Explanation of issues such as:
  - bug fixing, feature enhancement, refactoring, performance optimization, documentation

Provides screenshots or examples from the PITA project where relevant.

#### 4. Reflection (Individual Section)

Each student must provide a short reflection discussing:

- Knowledge gained from reviewing the technical papers
- Challenges encountered during the analysis
- Insights on the importance of software maintenance and evolution in real-world systems

#### 5. Conclusion (Group Section)

This section should summarize:

- Key insights from the technical paper analysis
- Key findings from the comparison with the PITA project
- The importance of applying effective software maintenance strategies

#### 6. References

All references must follow an APA referencing style.

### **3. REPORT FORMAT**

Students must ensure the report is properly formatted.

Writing format:

Font: Times New Roman

Font Size: 12

Spacing: 1.5 spacing

### **4. PLAGIARISM AND SIMILARITY CHECK**

Before submitting the final report, students must check the report using Turnitin.

1. The report must first be submitted to Turnitin to verify the similarity index.
2. The report is acceptable only if the similarity and AI similarity index is below 20%.
3. If the similarity exceeds 20%, the group must revise and resubmit the report until the similarity score meets the requirement.

### **SUBMISSION DEADLINES**

| <b>Activity</b>             | <b>Date</b>      |
|-----------------------------|------------------|
| Proposal Submission         | _16/4/2026_      |
| Technical Report Submission | _11/6/2026_      |
| Presentation                | _14 - 23/6/2026_ |